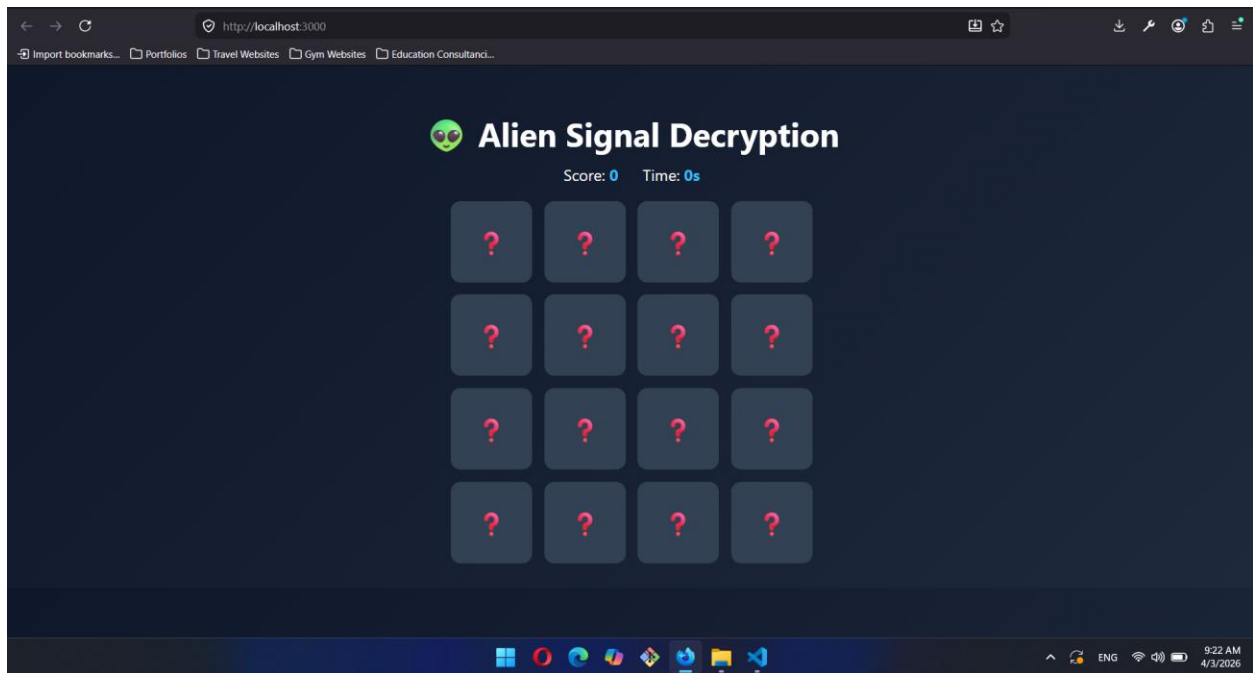


M Abdullah Roshaan

Section: A

Web-Lab #10

Task 1:



App.Js :

```
import React, { useState, useEffect } from "react";
import Card from "./Card";
import "./App.css";

function App() {
  const [cards, setCards] = useState([]);
  const [flippedCards, setFlippedCards] = useState([]);
  const [matchedCards, setMatchedCards] = useState([]);
  const [score, setScore] = useState(0);
  const [time, setTime] = useState(0);
  const [gameStarted, setGameStarted] = useState(false);

  useEffect(() => {
```

```

const signals = ["🀄", "🀅", "🀆", "🀇", "🀈", "🀉", "🀊", "🀋"];

const duplicated = [...signals, ...signals];

const shuffled = duplicated
  .map(signal => ({ id: Math.random(), signal }))
  .sort(() => Math.random() - 0.5);

setCards(shuffled);
}, []);

const handleCardClick = (card, index) => {
  if (
    flippedCards.length === 2 ||
    flippedCards.includes(index) ||
    matchedCards.includes(card.signal)
  ) {
    return;
  }

  if (!gameStarted) setGameStarted(true);

  setFlippedCards(prev => [...prev, index]);
};

useEffect(() => {
  if (flippedCards.length === 2) {
    const [first, second] = flippedCards;

    if (cards[first].signal === cards[second].signal) {
      setMatchedCards(prev => [...prev, cards[first].signal]);
      setScore(prev => prev + 100);
      setFlippedCards([]);
    } else {
      setScore(prev => prev - 10);

      setTimeout(() => {
        setFlippedCards([]);
      }, 800);
    }
  }
}, [flippedCards, cards]);

useEffect(() => {
  let timer;
  if (gameStarted) {
    timer = setInterval(() => setTime(prev => prev + 1), 1000);
  }
  return () => clearInterval(timer);
}, [gameStarted]);

```

```

useEffect(() => {
  if (matchedCards.length === 8) {
    alert(`🚀 Decryption Complete!\nScore: ${score}\nTime: ${time}s`);
  }
}, [matchedCards]);

return (
  <div className="app">
    <h1 className="title">👽 Alien Signal Decryption</h1>

    <div className="stats">
      <div>Score: <span>{score}</span></div>
      <div>Time: <span>{time}</span>s</div>
    </div>

    <div className="grid">
      {cards.map((card, index) => (
        <Card
          key={card.id}
          card={card}
          onClick={() => handleCardClick(card, index)}
          isFlipped={flippedCards.includes(index)}
          isMatched={matchedCards.includes(card.signal)}
        />
      ))}
    </div>
  </div>
);
}

export default App;

```

App.css:

```

body {
  margin: 0;
  font-family: "Segoe UI", sans-serif;
  background: linear-gradient(135deg, #0f172a, #1e293b);
  color: white;
}

.app {
  text-align: center;
  padding: 30px;
}

.title {

```

```

    font-size: 2.5rem;
    margin-bottom: 10px;
}

.stats {
    display: flex;
    justify-content: center;
    gap: 30px;
    margin-bottom: 20px;
    font-size: 1.2rem;
}

.stats span {
    color: #38bdf8;
    font-weight: bold;
}

.grid {
    display: grid;
    grid-template-columns: repeat(4, 100px);
    gap: 15px;
    justify-content: center;
}

```

Card.css:

```

.card {
    width: 100px;
    height: 100px;
    perspective: 1000px;
    cursor: pointer;
}

.card-inner {
    width: 100%;
    height: 100%;
    position: relative;
    transition: transform 0.5s;
    transform-style: preserve-3d;
}

.card.flipped .card-inner {
    transform: rotateY(180deg);
}

.card-front,
.card-back {
    position: absolute;
    width: 100%;
    height: 100%;
}

```

```

    border-radius: 12px;
    display: flex;
    justify-content: center;
    align-items: center;
    font-size: 30px;
    backface-visibility: hidden;
}

.card-front {
  background: #334155;
}

.card-back {
  background: #0ea5e9;
  transform: rotateY(180deg);
}

.card.matched .card-back {
  background: #22c55e;
  box-shadow: 0 0 15px #22c55e;
}

```

Card.js:

```

import React from "react";
import "./Card.css";

function Card({ card, onClick, isFlipped, isMatched }) {
  return (
    <div
      className={`card ${isFlipped || isMatched ? "flipped" : ""} ${isMatched
? "matched" : ""}`}
      onClick={onClick}
    >
      <div className="card-inner">
        <div className="card-front">?</div>
        <div className="card-back">{card.signal}</div>
      </div>
    </div>
  );
}

export default Card;

```

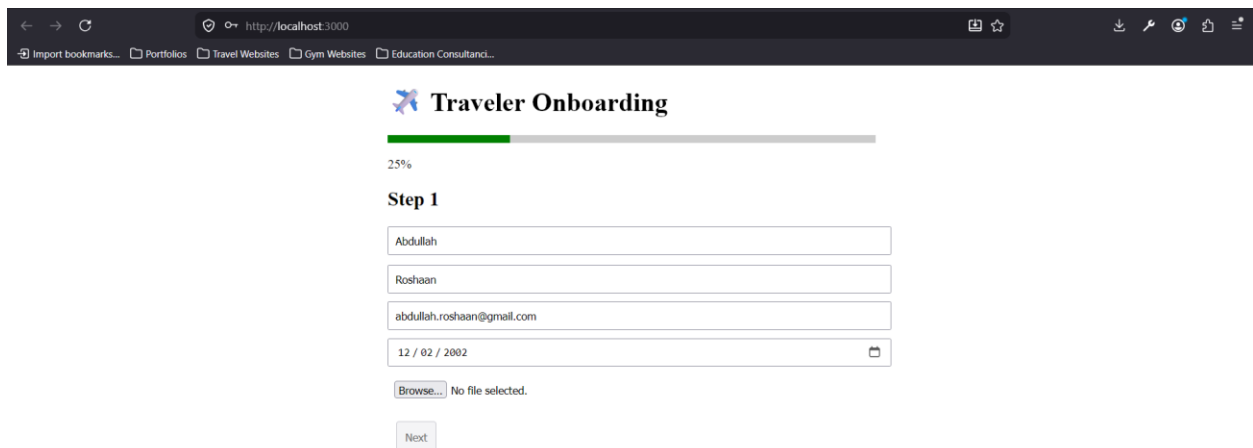
index.js:

```
import React from "react";

import ReactDOM from "react-dom/client";
import App from "./App";

const root = ReactDOM.createRoot(document.getElementById("root"));
root.render(
  <React.StrictMode>
    <App />
  </React.StrictMode>
);
```

Task #2:



Traveler Onboarding

25%

Step 1

Abdullah

Roshaan

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12 / 02 / 2002

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Next

Files in Codebase

Task #3:

